

2026-09-20 Baseline US Biopharma sample (1st round)

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Outline

- 1 Lessons from my last report and preview of the sample construction
- 2 Stage A: Obtain the broadest candidate employment spells
- 3 Stage B: Obtain the candidate new hires in US Biopharma industry
- 4 Stage C: Obtain the candidate LinkedIn job postings
- 5 Stage D: Match new hires to job postings

Lessons from my last report

Last report: [link](#).

- Bad news: Revelio's occupation classifications are noisy.
 - For example, when we focus on new hires (under occupation codes 17: Architecture and Engineering occupations and 19: Life, Physical, and Social Science occupations; between 2021 and 2023), there are a large share of astronomers (12.2%) and historians (11.8%).
 - This is caused mostly by the way Revelio assigns occupation codes for employment spells.
 - **Implications: It is necessary to construct sample based on users' self-reported job titles.**
- Good news: For inventors, most of their employment spells are under occupation codes 17 or 19.
 - The significantly missing occupations are mostly under occupation codes 15, e.g., software engineers, computer programmers. But anyway, they are never our main focus in the baseline Biopharma industry.
 - **Implications: This is good news because the broad occupation restriction based on 2-digit occupation codes can be justified by these inventors.**

Sample construction: Stage A and B

- Stage A: Obtain candidate employment spells (under occupation codes 17 and 19; starting between 2021 and 2023).
 - The main purpose is to reduce the sample size so that later steps can be done locally.¹
 - This is justified for our baseline Biopharma industry as most inventors' (who are our main focus) employment spells are classified under these 2 two-digit codes even though finer occupation codes are noisy.
- Stage B: Obtain candidate new hires based on job titles.
 - I keep only those researchers/scientists in the Biopharma industry in US.² After this step, the observation is at user-company level, where internal transfers are intentionally kept.
 - The sample is not our final sample of new hires because only a subset of them can be matched to job postings.

¹The development of the algorithm to construct sample based on job titles will be very difficult if the dataset is in the cloud.

²In this baseline sample, I will keep only scientists in US Biopharma industry because I currently don't have a good way to deal with job postings in other languages.

Sample construction: Stage C and D

- Stage C: Obtain candidate LinkedIn job postings.
 - Time restrictions: 12 months before the earliest new hire within the company, and 1 month before the last new hires within the company.
 - Occupation restrictions: First I restrict to job postings under codes 17 and 19, and then I apply the same job title restrictions as in Stage B to the job postings.
- Stage D: Match new hires to job postings.
 - Each new hire will be matched to multiple job postings (those postings that are 1-12 months before the new hire's start month).
 - Only those new hires that can be matched to job postings, and only those job postings that can be matched to new hires will be in our final baseline sample.
 - As I have said before, then we will weight these job postings so that one synthetic job posting is constructed for one new hire.

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Details in Stage A

- This stage creates a employment-spell-level dataset that is suitable to handle locally (instead of on the cloud).
- It starts from the user profile dataset, and applies only 3 restrictions.
 - ❶ The employment spell starts between [2021-01, 2024-01).³
 - ❷ The employment spell has non-missing position ID, user ID, company ID, country, and job title (raw or translated).
 - ❸ The ONET occupation codes start with 17 (Architecture and Engineering occupations) or 19 (Life, Physical, and Social Science occupations).

³This time frame is determined by (1) the sample period of the job postings and (2) the fact that we need to observe the new hires' later career paths.

Justifications for Stage A

- It is very hard to develop the algorithm to restrict job titles for a huge dataset that can only be accessed on the cloud.
 - There are 22,142,775 resulting employment spells after Stage A, which is of medium size and can be handle locally.
 - Otherwise, it is difficult to develop an algorithm to restrict job titles to those researchers/scientists/engineers of interest, and computation costs could be large.
- My last report has shown that most inventors' employment spells are under 17 and 19 (especially for the Biopharma industry).
 - Therefore, we won't lose many employment spells that are of interest but are classified under other occupation codes.
 - Note that software engineers are under code 15, so even though they are also important inventors, they are not included in the sample.

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Details in Stage B

- I start from the 22,142,775 employment spells obtained in Stage A and apply the following restrictions in sequence.
 - ① I keep employment spells in the United States.
 - ② I keep employment spells in Biopharma: the NAICS code either starts with 3254 (Pharmaceutical and Medicine Manufacturing) or is 541714 (Research and Development in Biotechnology).
 - ③ I remove employment spells whose job titles (after simple standardization⁴) indicate that they are outside the researchers/scientists/engineers of interest.
 - ④ For each user-company cell, I select the first employment spell that passes the job-title restrictions.

⁴Details for the standardization of job titles: [▶ Standardization of job titles](#)

Stage B sample size after each step

- After Stage A, there are 22,142,775 employment spells.
- Changes in the sample size after each step in Stage B:
 - United States: 4,619,058 employment spells.
 - US Biopharma: 292,141 employment spells.
 - After job title restrictions: 143,460 eligible employment spells (49.1% of the US Biopharma spells).
 - After selecting one spell for each user-company pair: 126,825 candidate new hires.
- The sample is not our final sample of new hires because only a subset of them can be matched to job postings. I will call this sample **candidate new hires**.

Which job titles are excluded

- I first exclude internships and similar student placements.
- I then exclude titles indicating work mainly in the following areas.
 - Data science; clinical operations; quality assurance, quality control, or compliance; and regulatory or medical affairs.
 - Operational engineering; sales, marketing, administration, and other commercial functions.
 - Safety and healthcare support; general software, IT, and business analytics; and non-employment positions such as retirement or board membership.
- I also exclude managers, directors, executives, founders, consultants, and similar titles when the title does not also contain an explicit research role.
 - A title such as “scientist manager” remains eligible, while a management title without a research role (e.g., “associate director”) is excluded.
 - I didn’t use the Revelio’s seniority variable in this step of job title restrictions. The exclusion rules depend only on users’ self-reported job titles.

Which job titles remain eligible

- A title remains eligible if it is not an internship and does not match any exclusion rule.
 - The rules recognize explicit research roles and functions such as scientist, chemist, engineer, clinical science, analytical science, process development, computational biology, drug discovery, and R&D.
 - However, the script does not require an eligible title to contain one of these terms.
- I keep several uncertain groups in the baseline sample but flag them for later review.
 - These include mixed researcher-manager titles, mixed quality-development titles, ambiguous technical titles, disagreements between raw and translated titles, and titles without a clear research role or function.

(Unpleasant) Summary statistics before and after job title restrictions

- Before job title restrictions, there are 292,141 employment spells (103,599 different job titles, 229,429 distinct users).
 - 21,973 employment spells (7.5%) are from USPTO-LinkedIn matched inventors. Among the distinct users, 16,385 of them (7.1%) are also inventors from the US patents database.
- After job title restrictions, there are 143,460 employment spells (39,581 standardized job titles, 115,973 distinct users).
 - 8,263 employment spells (5.8%) are from USPTO-LinkedIn matched inventors. Among the distinct users, 6,621 of them (5.7%) are also inventors from the US patents database.
- The unpleasant decrease in the inventor share comes from the fact that management roles (without recognized research-role words) are excluded from the sample under current job title exclusion rules (e.g., “chief scientific officer” or “director medicinal chemistry” or “vice president research and development”).
 - These high seniority users⁵ are more likely to show up in US patents database. This is somewhat tricky because titles like “associate director” may also be inventors but it is hard to tell from this title whether the user takes a researcher/scientist role.
 - All other job title exclusion rules increase inventor shares, but because most inventors in USPTO database are management roles, so its drop in inventor shares decreases the overall inventor shares. [▶ Details](#)
- I am currently working on a better version of job title exclusion rules, but because it is a bit time-consuming to finish the whole sample construction workflow, current report still uses these old rules.

⁵The median Revelio seniority score is 5 among management-excluded spells versus 2 among retained spells.

Job titles in the excluded employment spells



- Before job title restrictions, there are 292,141 employment spells in US Biopharma industry. After job title restrictions, there are 143,460 employment spells.
- This word cloud plots the top 500 job titles among the 148,681 (= 292,141 - 143,460) employment spells who have non-eligible job titles.

Job titles in the eligible employment spells

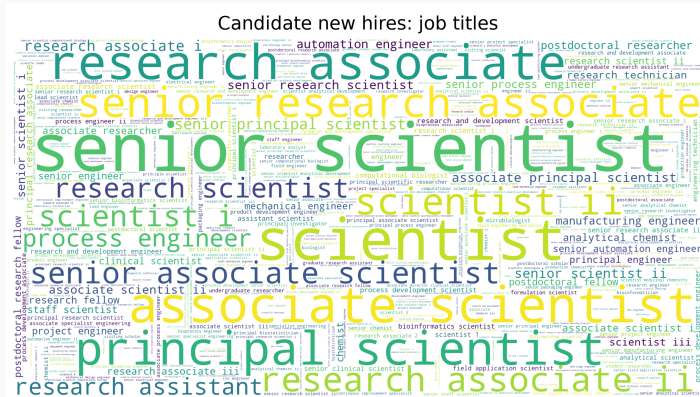


- Before job title restrictions, there are 292,141 employment spells in US Biopharma industry. After job title restrictions, there are 143,460 employment spells.
- This word cloud plots the top 500 job titles among the 143,460 employment spells who have non-eligible job titles.

From employment-spell-level to user-company-level

- I form one observation for each user-company pair that has at least one eligible employment spell.
- When a user-company pair has more than one eligible spell, I select one intact source spell using the following order.
 - ① The earliest start date.
 - ② If there is a tie, the lowest observed seniority value.
 - ③ If there is still a tie, the lowest position number.
- This gives the first eligible spell observed for the user-company pair during 2021–2023.
 - It is not necessarily the worker's first-ever spell at the company.
 - Internal transfers are intentionally kept because internal transfers vs. external hires could be meaningfully different hiring practices.

Job titles in the final sample of candidate new hires (user-company level)



- There are 126,825 candidate new hires (the sample is at user-company level) after Stage B. This word cloud plots the top 500 job titles among these 126,825 new hires. (There are 35,876 distinct job titles.)
 - Top 10 job titles: senior scientist (6.3%), scientist (4.6%), associate scientist (3.4%), research associate (3.2%), principal scientist (2.9%), senior research associate (2.4%), scientist ii (1.9%), research associate ii (1.6%), senior associate scientist (1.4%), scientist i (1.4%).
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Details in Stage C

- Stage C uses the 126,825 candidate new hires from Stage B to obtain a sample of candidate LinkedIn job postings that may be matched to them.
- I divide this stage into two steps.
 - ① Stage C1 constructs a posting-date window for each company in the Stage B sample.
 - ② Stage C2 extracts LinkedIn postings for these companies and applies restrictions on country, occupation code, posting date, and job title.
- A posting retained in Stage C is only a candidate posting, not necessarily in the final sample.
 - Stage C does not yet match a posting to a particular new hire.
 - Recall the ghost job posting issue I have raised before ([link](#)): “The ratio of hires per job posting in the US has fallen from 0.75 in 2018 to below 0.5 in 2023, meaning fewer than one in two US job postings now lead to an actual hire.”
 - If a job posting cannot be matched to any new hire, it will not be in our final sample.

Stage C1: Company-specific posting-date windows

- I start from all 126,825 candidate new hires selected in Stage B and construct one posting-date window for each company.
 - The Stage B sample contains 5,939 companies, so Stage C1 produces 5,939 company windows.
- For each company, the lower bound is 12 calendar months before its earliest candidate new hire, and the upper bound is 1 calendar month before its latest candidate new hire.
 - In Stage D, I will match all job postings within 1-12 months before a new hire's start month to the corresponding new hire.
 - As previously discussed, I will future weight these matched job postings to construct one synthetic job posting for each new hire.

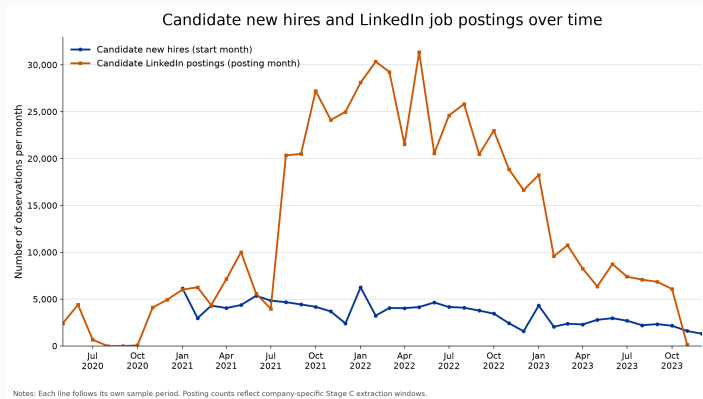
Stage C2: Initial restrictions on LinkedIn postings

- I start from the full LinkedIn posting table and first keep all postings from the 5,939 companies.
- I then apply the following restrictions in sequence.
 - ① I require a non-missing posting ID and at least non-missing job title (raw or translated).
 - ② I keep postings in the United States whose ONET occupation code starts with 17 or 19.
 - ③ I keep postings whose post date is inside the company-specific posting window (created in Stage C1).
 - ④ I apply the same job-title restrictions used in Stage B and keep eligible postings.
- I do not additionally restrict the posting's reported industry because the company universe already comes from the US Biopharma new hires selected in Stage B.

Stage C2: Restrictions on posting dates and job titles

- The posting date, rather than the removal date, determines whether a posting falls inside the company window.
 - A missing or invalid posting date cannot pass the date restriction.
- I standardize the raw and translated titles and apply the same job title restrictions as in Stage B.
 - A posting is excluded if either title matches an exclusion rule.

Time series: Candidate new hires and candidate LinkedIn job postings



- After Stage B, there are 126,825 candidate new hires.
- After Stage C, there are 557,567 candidate LinkedIn job postings.

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Details in Stage D

- Stage D starts from the 126,825 candidate new hires in Stage B and the 557,567 candidate LinkedIn job postings in Stage C.
- For each candidate new hire, I construct a hire-specific posting window.
 - The lower bound is 12 calendar months (inclusive) before the recorded employment start date.
 - The upper bound is 1 calendar month (inclusive) before the recorded employment start date.
- I link the new hire to every candidate posting from the same company whose posting date falls inside this window.

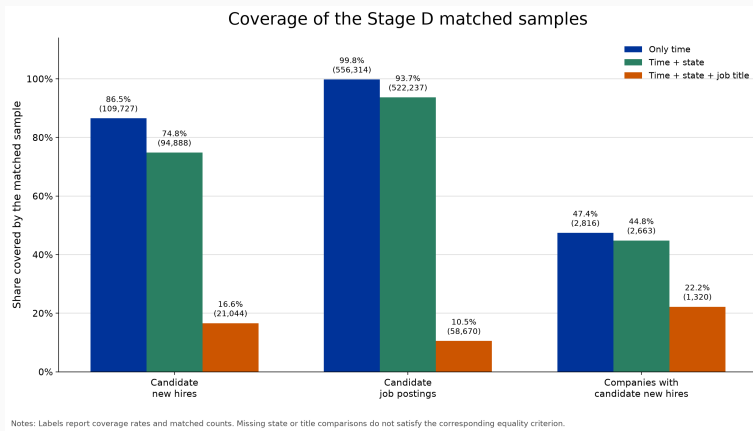
Interpretations of the many-to-many match in Stage D

- One new hire can be linked to multiple prior job postings.
 - The script does not select one “winning” posting for a new hire.
 - As discussed in a previous meeting, we will create one synthetic job posting by weighting the pool of matched job postings.
 - A link identifies a posting that may describe the recruitment environment faced by a later new hire at the same company.
 - It does not establish that the company used that posting to hire that worker.
 - It does not establish that a posting represents one unique vacancy.
- One job posting can also be linked to multiple later new hires.
 - An unmatched posting does not necessarily represent a failed vacancy (the new hire may not be a LinkedIn user), and an unmatched new hire does not have zero recruitment requirements.

Three criteria for the matching process

- I consider a new hire is matched to a job posting with 3 different criteria.
 - The first case is "only time" criterion. A company's new hire is matched to the same company's job postings as long as the posting month is 1 to 12 months (inclusive) before the new hire's start month.
 - The second case is "time + state" criterion. Beyond the previous criterion, I also require the state of the new hire to be the same as the state of the job postings.
 - The second case is "time + state + job title" criterion. Beyond the previous 2 criteria (time + state), I also require the normalized job title of the new hire to be the same as the normalized job title of the job postings.
- As we will see later in the summary visualization, the "only time" criterion will lead to many "false positive" links between new hires and job postings.
 - My current time window (i.e., $[-12, -1]$) is wide enough. Audoly et al. (2026) uses a 6 month period to construct the matched samples of new hires and job postings with Norwegian admin data. [Details](#).
 - Personally, I prefer the "time + state + job title" criterion, even though it reduces the sample size by a lot.

Coverage rates across 3 criteria



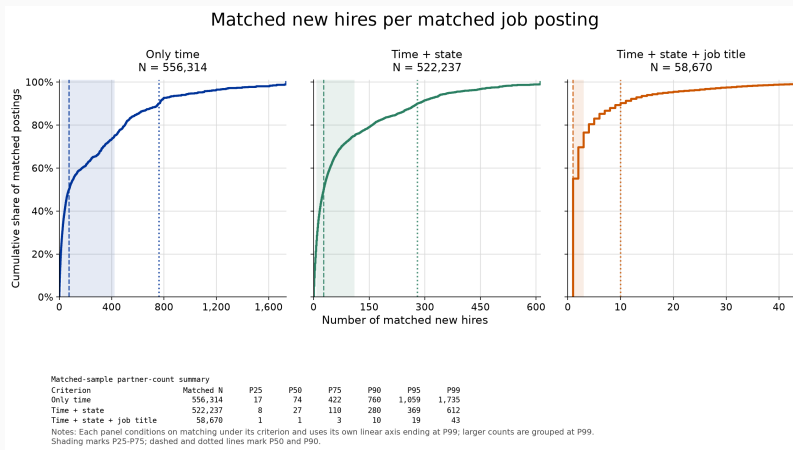
- There are 126,825 candidate new hires, and 557,567 candidate LinkedIn job postings.
- Audoly et al. (2026) uses a very conservative method to construct the matched samples of new hires and job postings with Norwegian admin data, their matched job postings are around 11% of the full job postings data – very similar to our current 10.5%.
- I will talk more about this paper in the other report because it is really relevant and I highly recommend it.

The number of matched job postings per matched new hire



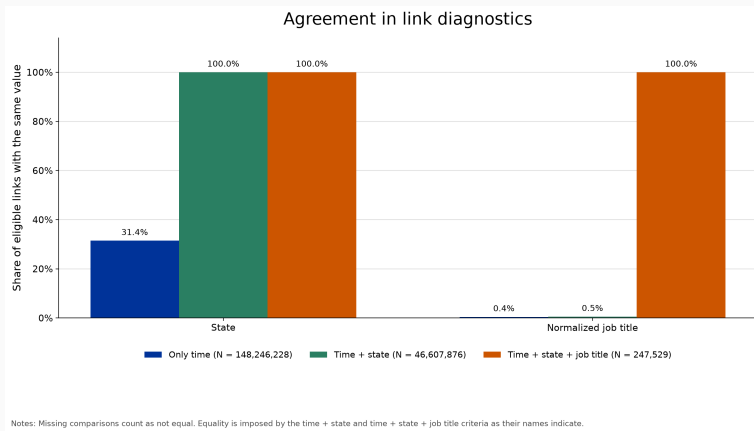
- There are 126,825 candidate new hires, and 557,567 candidate LinkedIn job postings.
- Under "time + state + job title" criterion, even though there are only 21,044 matched new hires, the distribution of the number of matched job postings is quite reasonable.

The number of matched new hires per matched job posting



- There are 126,825 candidate new hires, and 557,567 candidate LinkedIn job postings.
- Under "time + state + job title" criterion, even though there are only 58,670 matched job postings, the distribution of the number of matched new hires is quite reasonable.

Share of matched sample with the same state/job titles across 3 criteria



- There are 126,825 candidate new hires, and 557,567 candidate LinkedIn job postings. Note that the reported N in the legend is the number of links (between matched new hires and job postings).
- Under "time + state" criterion, there are only 0.5% of links with the same (normalized) job titles.

Thoughts on the matched sample of new hires and job postings

- We definitely don't want a lot of "false positive" matches between new hires and job postings.
- The tradeoff is that when we impose more restrictions to the matches (from "only time" to "time + state + job title"), we get more confident matches, but we lose a lot of observations.
 - After Stage B and C, we have 126,825 candidate new hires, 557,567 candidate LinkedIn job postings, and 5,939 companies.
 - Under the most strict criterion ("time + state + job title"), there are only 21,044 matched new hires, 58,670 matched job postings, and 1,320 companies in the baseline US Biopharma industry.
 - We could lost additional observations in the future when we focus only on the meaningful job postings in the sense that we can construct variables from the advertisement texts.
- A practical solution is to extend current baseline US Biopharma companies to global Biopharma companies.

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⑥ Appendix: Additional contents

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Standardization of job titles

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- I use both the raw self-reported title and the translated title provided by Revelio.
 - The raw title is used as the main standardized title when it is available; otherwise, I use the translated title.
 - However, the exclusion rules are applied to both versions. A spell is excluded if either version matches an exclusion rule.
- Before applying the rules, I make simple changes that do not alter the meaning of the title.
 - I use lower case, remove punctuation and extra spaces, and expand explicit abbreviations such as “sr” to “senior” and “r&d” to “research and development.”
 - I keep word order, scientific qualifiers, and seniority. I do not use stemming or fuzzy matching.
 - Ambiguous abbreviations such as “RA,” “AD,” and “PD” are not expanded.

Inventor shares before and after job title restrictions

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Exclusion rule	Change in inventor shares (employment spell level)	Change in inventor shares (user level)
Internship	+0.5503	+0.5558
Missing title	+0.0008	+0.0009
Data science	+0.0044	+0.0026
Clinical operations	+0.4807	+0.4684
Quality/compliance	+0.4841	+0.4843
Regulatory/medical affairs	+0.0587	+0.0587
Operational engineering	+0.1528	+0.1478
Commercial/administration	+0.0045	+0.0034
Safety/healthcare	+0.0210	+0.0246
IT/business analytics	+0.0084	+0.0086
Management without research role	-3.8455	-3.3975

Notes. Values in the cells report changes in matched-inventor shares in percentage points when the indicated exclusion rule is added, **holding all other rules active**. Positive values indicate an increase in inventor shares.

Audoly et al. (2026) workflow to construct matched samples of new hires and job postings

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Audoly et al. (2026) also matches new hires to job postings using Norwegian admin data.

- 1 First, they search for workers starting employment: (1) at the same establishment, (2) at the same four-digit occupation, and (3) within the six months following the posting date.
- 2 Second, they restrict links to establishment-occupation cells with only one hire in the following six months.
- 3 Third, they exclude ads posted on the same date in the same establishment-occupation cell.
- 4 Fourth, they remove hires of workers who previously had an employment spell with that employer. Once an employment spell is linked to a posting, they do not allow that same spell to be linked to another posting.

They have a very conservative match process, and the matched sample of job postings consist of around 11% of all postings in the full job postings dataset.

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6 Appendix: Additional contents

7 Appendix: References

References I

Audoly, Richard, Manudeep Bhuller, and Tore Adam Reiremo (2026). “The Pay and Non-Pay Content of Job Ads”. *Working Paper*.